IP Addressing

What is an IP Address?

- An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network.
- There are two versions of IP that currently coexist in the global Internet: IP version 4 (IPV4) and IP version 6 (IPV6).

IPV4 Address:

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- **32 Bit** of Logical Address
- Provided by **IANA.**
- It is having 4 Blocks separated by "**.**" called **4 Octets**
- **Range** of an octet **is 0-255** and size of **8 Bit.**
- Example: **192.168.39.240**

  11000000.10101000.00100111.11110000 (Binary)
- Minimum Value of IP Address: **0.0.0.0**

  00000000.00000000.00000000.00000000 (Binary)
- Maximum Value IP Address: **255.255.255.255**

  11111111.11111111.11111111.11111111 (Binary)

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Classes of IP: -

- **Class A: Range 0.0.0.0 to 126.255.255.255**
- **Class B: Range 128.0.0.0 to 191.255.255.255**
- **Class C: Range 192.0.0.0 to 223.255.255.255**
- **Class D: Range 224.0.0.0 to 239.255.255.255 (reserverd)**
- **Class E: Range 240.0.0.0 to 255.255.255.255**

- **127.0.0.0 to 127.255.255.255 is reserved for the loopback IP Addresses which called the internal address of device.**

Example: So there can be (and is) network traffic just between applications running on the machine.

Example:

Q. What is class of IP: 103.108.5.56

Ans. Look at first Octet 103.108.5.56

-Range of Class A is 0-127

-IP Address belong to Class A.

Q. What is class of IP: 192.168.5.56

Ans. Look at first Octet 192.108.5.56

-Range of Class C is as per first octet is 192-223.

-IP Address Belong to Class C.

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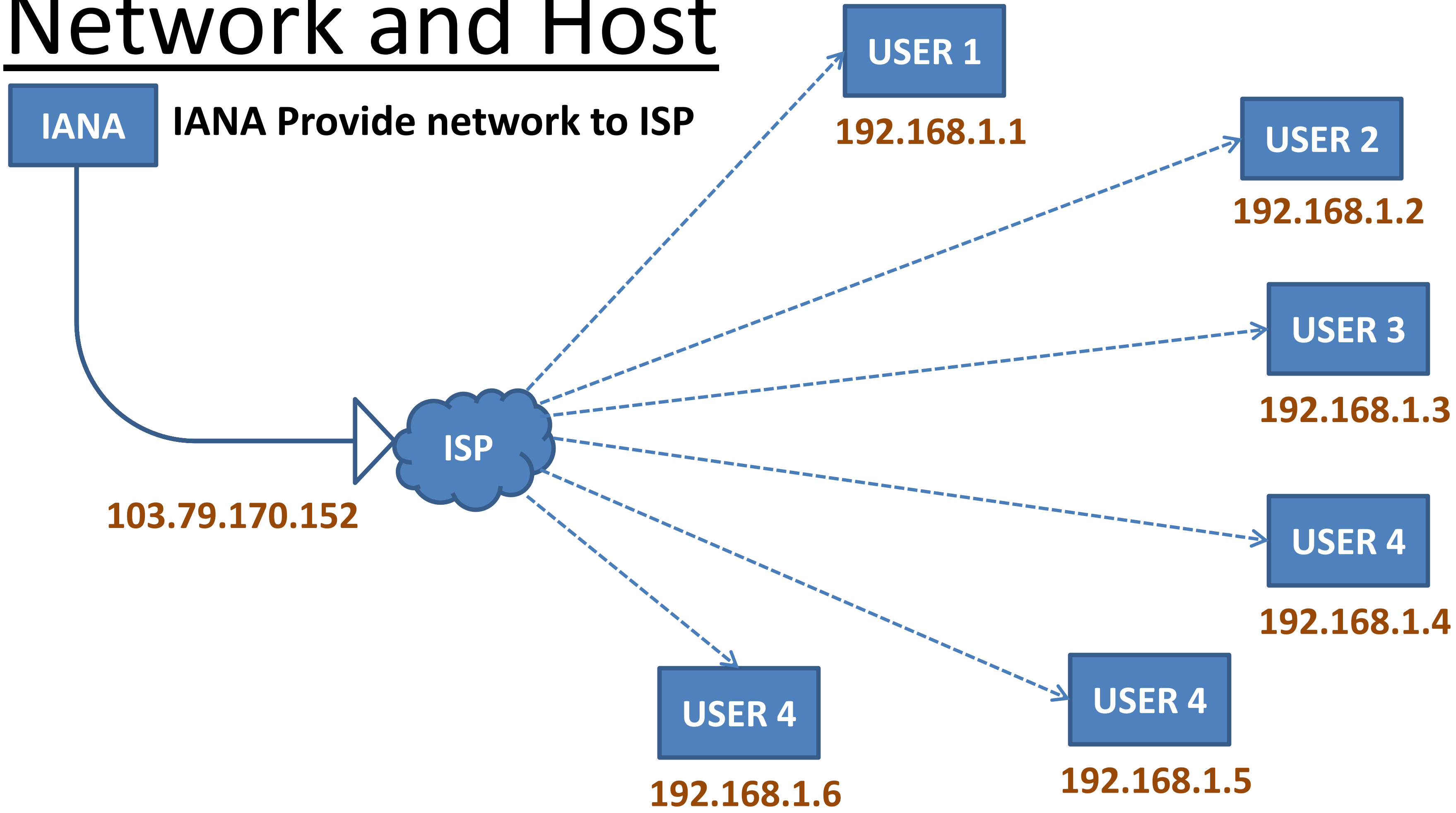
Network and Host



Network & Host Format for Classes :

- **Class A: N.H.H.H or**

NNNNNNNN.HHHHHHHH.HHHHHHHH.HHHHHHHH

Subnet Mask**: 255.0.0.0** which is Equivalent to

11111111.00000000.00000000.00000000

- **Class B: N.N.H.H or**

NNNNNNNN.NNNNNNNN.HHHHHHHH.HHHHHHHH

Subnet Mask**: 255.255.0.0** which is Equivalent to

11111111.11111111.00000000.00000000

- **Class C: N.N.N.H or**

NNNNNNNN.NNNNNNNN. NNNNNNNN.HHHHHHHH

Subnet Mask**: 255.255.255.0** which is Equivalent to

11111111.11111111.11111111.00000000

IPV4 Address: Host and Networks:

- **Class A: Range 0.0.0.0 to 127.255.255.255**

Networks = 126,

Host = 256*256*256=16,777,216 Host/Networks

- **Class B: Range 128.0.0.0 to 191.255.255.255**

Networks = 64*256=16,364,

Host = 256*256=65536 host/Networks

- **Class C: Range 192.0.0.0 to 223.255.255.255**

Networks = 32*256*256=2,097,152,

Host = 256 host/Networks

Examples:

Email: Need 60 IP Addresses for a Department in a company. Which class of IP will be alloted?

Ans: Class C, because Number of Host in Class C is 256 and this is nearest to 60.

Email: Need 1000 IP Addresses for a Department in a company. Which class of IP will be alloted?

Ans: Class B, because Number of Host in Class B is 65536 and this is Suitable for 1000.

Note* If we use class C IP Address, It will not fulfill the requirement.

How to Allot 60 IP Address:

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Let us Assume network is 199.99.9.0

The IP Address Allotted will be:

199.99.9.0- Network IP

199.99.9.1

199.99.9.2

- .
- Network
- Is Fixed

199.99.9.254

199.99.9.255 – Broadcast IP

Total Valid IP=256-2=254, Wastage: 254-60=194

Private IP Address:

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A private IP address is a non-routable address of your device assigned by the network router. Every device connected to the same network has a private IP or network address. An IP address allows devices connected over the same network to communicate.

Ranges of Private IP Address:

- **Class A: Range 10.0.0.0 to 10.255.255.255**
- **Class B: Range 172.16.0.0 to 172.31.255.255**
- **Class C: Range 192.168.0.0 to 192.168.255.255**

Private IP Address: Host and Networks:

- **Class A: Range 10.0.0.0 to 10.255.255.255**

Networks = 1,

Host = 256*256*256=16,777,216/Networks

- **Class B: Range 172.16.0.0 to 172.31.255.255**

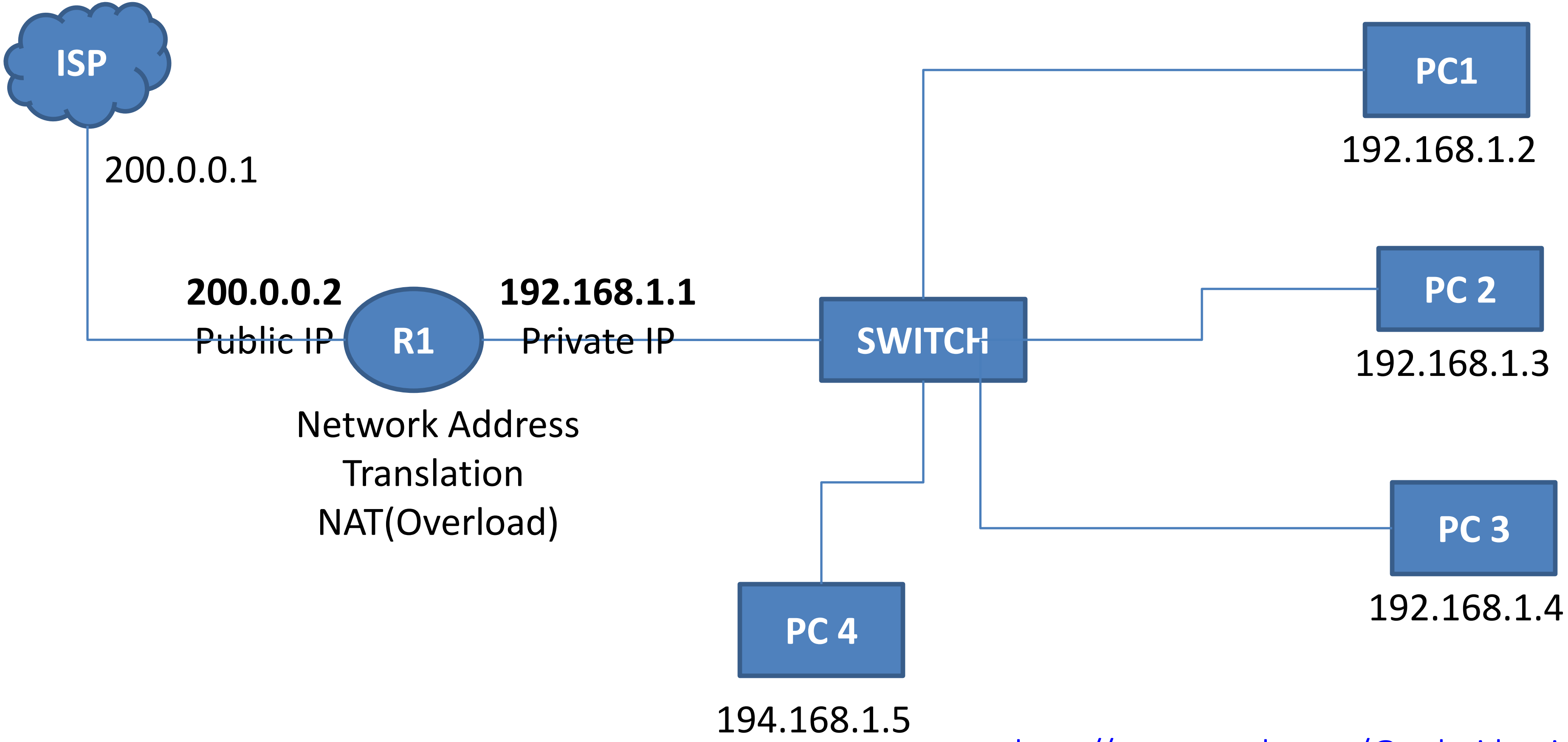
Networks = 16,

Host = 256*256=65536/Networks

- **Class C: Range 192.168.0.0 to 192.168.255.255**

Networks = 256, Host = 256/Networks

How to Convert Public to Private IP:



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